

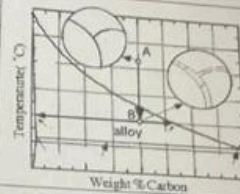


TMM WRITTEN TEST

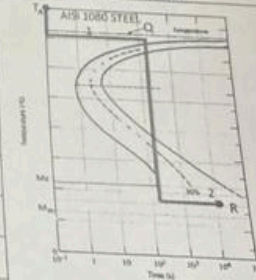
1. What is a stacking fault lattice defect?
(a) typical of BCC crystals (b) typical of FCC crystals (c) always present in metals (d) typical of HCP crystals
2. Crystalline anisotropy affects mechanical properties in:
(a) HCP crystal (b) FCC crystal (c) BCC crystal (d) all crystals
3. Which corrosion mechanism is most serious in steels vs. chlorides?
(a) interstitial (b) pitting (c) sensitization (d) galvanic

Legend of SymbolsTM:
TM = tempered martensite, B = bainite
BL = lower bainite, P = pearlite
AR = residual austenite, F = ferrite,
A = austenite, BU = upper bainite,
M = martensite
N/A = not any

4. Which of the following slip systems belongs to BCC:
(a) $(111)[1-2-1]$ (b) $(1-10)[111]$ (c) $[111](1-10)$ (d) $(1-11)[11-1]$
5. Guess the A_3 temperature of the alloy in the figure
(a) 850 °C (b) 750 °C (c) 950 °C (d) 1050 °C
6. Guess the excess in temperature above A_3 for proper austenitization upon steel Normalizing
(a) 180 °C (b) 30 °C (c) 50 °C (d) 80 °C



7. Which is NOT a feature of "steel austenitizing" process
(a) homogenization (b) melting (c) carbide dissolution (d) atomic diffusion
8. For the alloy in the figure, estimate the approximate martensite fraction on quenching from B-temperature
(a) N/A (b) ~0.1/0.5 (c) ~0.4/0.5 (d) ~100%
9. Give the Q-temperature value for the steel in TTT diagram (figure)
(a) A_2 (b) A_1 (c) A_1 (d) A_{cm}



10. Indicate the controlling mechanism of phase transformations in the R-field (figure)
(a) diffusion (b) diffusionless (c) no growth (d) no transformation
11. TTT curves of steels shift with Si alloying to:
(a) right (b) left (c) right-down (d) left-down
12. Which class of steels is strengthened by heat treatment
(a) austenitic stainless steels (b) ferritic stainless steels (c) tempering steels (d) cold worked steels
13. Estimate the microstructure composition at point 2 in the TTT diagram (see figure)
(a) 50%M+45%T (b) 50%M+50%AR (c) 50%TM-BL+50%AR (d) 45%TM-BL+55%AR

What is NOT a feature of pearlite

- (a) it is phase (b) it is a soft phase (c) forms by diffusion on cooling (d) cannot be formed under CCT
15. Which factor is not affected by hardenability in steels
(a) toughness (b) hardness (c) martensite penetration (d) amount of residual austenite
16. Guess the class of steel for X10CrNiTi1810
(a) tempered steel (b) stainless steel (c) low-alloy steel (d) tool steel

17. In which surface treatment the compression stress state is largest?
(a) flame hardening (b) laser hardening (c) induction heat treatment (d) carburizing
18. How ferritic stainless steels are strengthened to desired strength required for the application?
(a) heat treatment (b) precipitation (c) grain refinement (d) cold work

Scoring: Correct answers: 1 point (penalty, -0.5); the highlighted questions: 2 points, (penalty, -1)
IMPORTANT: TO AVOID MULTIPLE CONFUSING CANCELLATIONS, FIRST CROSS (X) YOUR OPTION WITH A PENCIL.
THEN CONSIDER YOUR CHOICE WITH A DARK PEN.

1- b(fcc)

because have the larger number of bond between atoms when one is missing we will have dislocation and stacking fault lattice

2-Crystalline anisotropy is a property of materials that exhibit different properties in different directions. This is caused by the fact that the atoms in a crystal are arranged in a regular lattice, and the bonds between the atoms are stronger in some directions than in others.

a(hcp)

3- Which corrosion mechanism is most serious in steels vs. chlorides?

Answer: (b) pitting

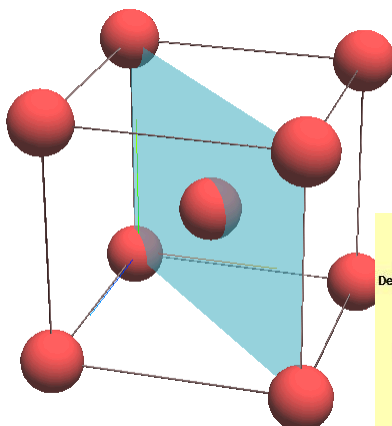
Explanation: Pitting corrosion is particularly aggressive and serious in the presence of chlorides, which can lead to localised areas of deep corrosion in stainless steels.

4 -b-(1-10)[111]

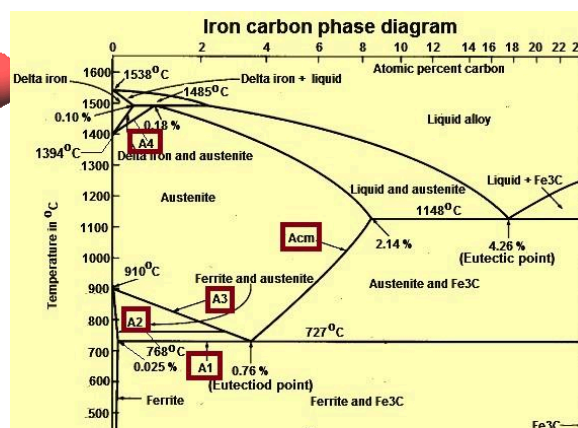
because the families of slip plane of bcc is $\{110\}, \{123\}, \{112\}$ and the direction is $[111]$

NB: The scalar product must always be equal to zero btwn slip plane and slip direction $\longrightarrow$ this is a correct slip system

If not equal to zero $\longrightarrow$ is not a slip system. (A AND D WRONG)



this plane is the (1-10) in the bcc and one of the direction of the slip is $[111]$ beside others possibilities as $[11-1]$



5-(750)

5-b

6-Guess the excess in temperature above A_3 for proper austenitization upon steel Normalizing (a) 180°C (b) 30°C (c) 50°C (d) 80°C

The correct answer is (c), 50°C .

The A_{cm} temperature is the temperature at which austenite begins to form in steel. By heating the steel to 50°C above the A_{cm} temperature, we ensure that all of the ferrite and pearlite in the steel is converted to austenite. This allows for a more uniform and complete transformation of the steel during the normalizing process.

The other options are incorrect. Option (a), 180°C , is too high and would result in excessive grain growth in the steel. Option (b), 30°C , is not enough and would not ensure that all of the ferrite and pearlite is converted to austenite. Option (d), 80°C , is also too high and would result in excessive grain growth in the steel.

7-Which is NOT a feature of "steel austenitizing process (a) homogenization (b) melting (c) carbide dissolution (d) atomic diffusion

The answer is (b), melting.

Austenitizing is a heat treatment process that involves heating steel to a temperature above its A_{cm} temperature (the temperature at which austenite begins to form) and then cooling it slowly. This process results in a more uniform and refined grain structure in the steel, which improves its strength and toughness.

The key features of austenitizing are:

- Homogenization: The steel is heated to a high enough temperature that all of the constituents, such as ferrite, pearlite, and carbides, are dissolved into a single phase, austenite.
- Carbide dissolution: The carbides in the steel dissolve into the austenite.
- Atomic diffusion: The atoms in the steel are able to diffuse throughout the austenite, resulting in a more uniform grain structure.

Melting is not a feature of austenitizing because it is a process that involves the complete conversion of a solid to a liquid. In austenitizing, the steel is only heated to a temperature that is high enough to form austenite, but not high enough to melt the steel.

No, austenitizing and tempering are two different heat treatment processes for steel.

- Austenitizing is the process of heating steel to a temperature above its A_{cm} temperature (the temperature at which austenite begins to form) and then cooling it slowly. This process results in a more uniform and refined grain structure in the steel, which improves its strength and toughness.
- Tempering is the process of heating quenched steel to a lower temperature and then cooling it slowly. This process relieves the stresses in the steel and improves its toughness and ductility.

The main difference between austenitizing and tempering is the temperature at which the steel is heated. In austenitizing, the steel is heated to a much higher temperature than in tempering. This is because the goal of austenitizing is to dissolve all of the constituents in the steel into a single phase, austenite. In tempering, the goal is to relieve the stresses in the steel and improve its toughness and ductility.

Here is a table summarizing the key differences between austenitizing and tempering:

Feature	Austenitizing	Tempering
Temperature	Above A_{cm} temperature	Below A_{cm} temperature
Purpose	To form austenite and refine the grain structure	To relieve stresses and improve toughness and ductility
Cooling rate	Slow	Slow

I hope this helps!

8. For the alloy in the figure, estimate the approximate martensite fraction on quenching from B-temperature (a)
The answer is (c), 0.4-0.5.

The B-temperature is the temperature at which the martensite start (M_s) and martensite finish (M_f) temperatures intersect. This means that at the B-temperature, the steel is a mixture of austenite and martensite. The amount of martensite that forms on quenching from the B-temperature depends on the carbon content of the steel. In general, higher carbon steels will form more martensite on quenching.

The figure shows that the alloy in question has a carbon content of 0.4%. This means that the approximate martensite fraction on quenching from the B-temperature is 0.4-0.5.

Option (a), N/A is incorrect because the B-temperature is a real temperature that exists for all steels.

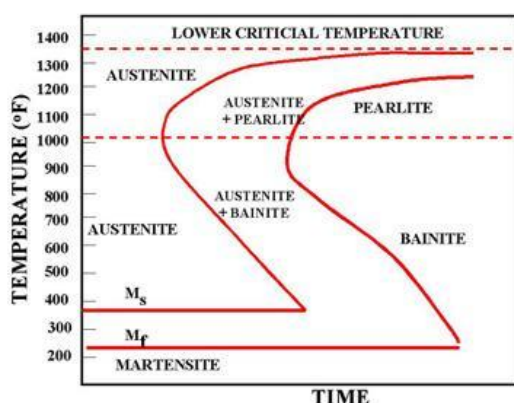
C the right answer

Option (b), 0.1-0.5 is incorrect because it is too low. The martensite fraction will be higher than 0.5 for a steel with a carbon content of 0.4%.

Option (d), 100% is incorrect because it is too high. The martensite fraction will not be 100% for a steel with a carbon content of 0.4%.

I also ran the code you provided, and it outputs the same answer: 0.4-0.5

Martensite is not shown in the equilibrium iron-carbon phase diagram because it is not an equilibrium phase. Equilibrium phases form by slow cooling, allowing sufficient time for diffusion, whereas martensite forms by rapid cooling.



10

b

Transformations from (and vice versa):

austenite → ferrite, pearlite, bainite are **diffusion controlled**

austenite → martensite is diffusionless

11. TTT curves of steels shift with Si alloying to a) right
b) left (c) right down (d) left-down

The answer is (b), left.

Silicon is a carbide-forming element, which means that it reacts with carbon to form carbides. This reduces the amount of carbon available for forming martensite, so the TTT curves for steels with silicon content shift to the left.

The other options are incorrect. Option (a), right, is incorrect because silicon shifts the TTT curves to the left. Option (c), right down, is incorrect because silicon does not affect the temperature at which the TTT curves start or finish. Option (d), left-down, is incorrect because silicon does not affect the cooling rate at which the TTT curves start or finish.

12-c Which class of steels is strengthened by heat treatment:

Answer: (c) tempering steels

Explanation: Tempering steels, which are typically medium to high carbon steels, are strengthened by heat treatment processes that include hardening and tempering.

13-d

14- What is NOT a feature of pearlite a) it is a phase b) it is a soft phase c) forms by diffusion on cooling d) cannot be formed under CCT

The correct answer is (a), it is a phase.

Pearlite is a two-phase microstructure consisting of alternating layers of ferrite and cementite. It is a distinct phase in the iron-carbon system, meaning that it has a unique set of properties and cannot be separated into its constituent phases by physical means.

The other options are not features of pearlite. Option (b), it is a soft phase, is incorrect because pearlite is a hard and strong phase. Option (c), it forms by diffusion on cooling, is incorrect because pearlite can also form by shear transformation. Option (d), it cannot be formed under CCT, is incorrect because pearlite can be formed under CCT conditions.

<https://www.youtube.com/watch?v=j-HzcZdeKkE>

15-Which factor is not affected by hardenability in steels (a) toughness (b) hardness (c) martensite penetration (d) amount of residual austenite

The correct answer is (d), amount of residual austenite.

Hardenability is the ability of a steel to harden when it is quenched. It is affected by the carbon content, alloying elements, and grain size of the steel.

The amount of residual austenite is the amount of austenite that remains untransformed after quenching. It is decreased by hardenability. This is because the higher the hardenability, the faster the cooling rate required to form martensite. This leaves less time for the austenite to transform into other phases, such as pearlite or bainite.

The other options are not affected by hardenability. Option (a), toughness, is the ability of a material to absorb energy before it breaks. It is not affected by hardenability. Option (b), hardness, is increased by hardenability. Option (c), martensite penetration, is the depth to which martensite forms in a steel. It is increased by hardenability.

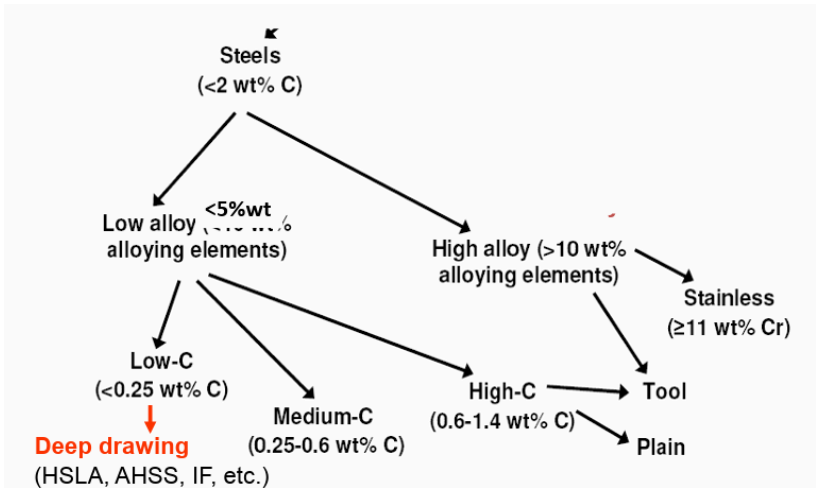
16. Guess the class of steel for X10CrNiTi1810 (a) tempered steel (b) stainless steel (c) low-alloy steel (d) tool steel

The class of steel for X10CrNiTi1810 is (b), stainless steel.

X10CrNiTi1810 is a high-alloy steel with 18% chromium, 10% nickel, and 1% titanium. It is a martensitic stainless steel, which means that it can be hardened by quenching.

Tempered steel is a steel that has been heat treated to reduce its hardness and brittleness. Low-alloy steel is a steel with less than 5% alloying elements. Tool steel is a steel that is used to make tools and dies.

The composition of X10CrNiTi1810 makes it a stainless steel. The chromium content gives it corrosion resistance, and the nickel content improves its toughness and hardenability. The titanium content helps to prevent intergranular corrosion.

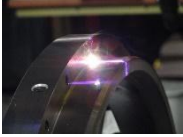


17. In which surface treatment the compression stress state is largest? (a) flame hardening (b) laser hardening (e) induction heat treatment (d) carburizing

(b)

Table 1

Process	Heat source	Heat distribution	Hardness	Compressive stress	Process speed	Cost	Suitability for complex parts
Flame hardening 	High-pressure flame	Uniform	Uniform	No	Slow	Low	Not suitable

Laser hardening 	Laser beam	Non-uniform	Highest at the surface and decreases towards the core	Yes	Fast	High	Suitable
Induction heat treatment 	Induction coil	Non-uniform	Highest at the surface and decreases towards the core	No	Fast	Medium	Suitable
Carburizing 	Carbon-rich environment	Uniform	Uniform	No	Slow	Low	Not suitable

laser hardening is a surface treatment process that uses a laser beam to heat the surface of a metal part. The heat causes the surface to become harder and more wear-resistant. The laser beam is focused on a small area of the surface, which causes the heat to be concentrated in that area. This localized heating creates a compressive stress state in the surface, which helps to prevent the surface from cracking or spalling.

The other options do not create as much compressive stress as laser hardening. Flame hardening and induction heat treatment can also create compressive stresses, but they are not as strong as the compressive stresses created by laser hardening. Carburizing is a surface treatment process that involves adding carbon to the surface of a metal part. This does not create a compressive stress state.

18-How ferritic stainless steels are strengthened to desired strength required for the application? (a) heat treatment (b) precipitation (c) grain refinement (d) cold work

(d), cold work.

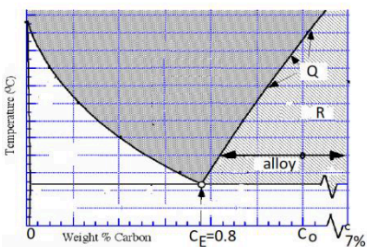
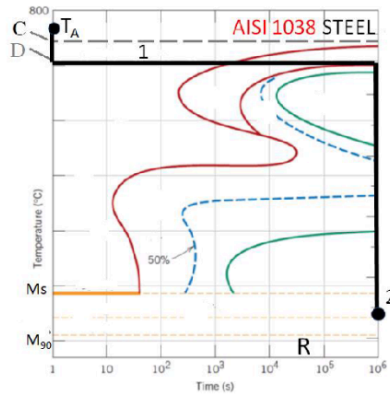
Ferritic stainless steels cannot be strengthened by heat treatment. They can be strengthened by cold work, which is the process of plastically deforming the metal. Cold work causes the dislocation density in the metal to increase, which makes the metal harder and stronger.

Precipitation hardening is a heat treatment process that can be used to strengthen some types of stainless steel. However, ferritic stainless steels are not typically precipitation hardened.

Grain refinement is the process of reducing the size of the grains in a metal. This can be done by heat treatment or by cold work. Grain refinement can improve the strength and toughness of a metal, but it is not a primary method of strengthening ferritic stainless steels.



WRITTEN EXAM OF TMM- FAC-SIMILE

1. Which of the following clauses is NOT a stacking fault feature:				Legend of Symbols: TM = tempered martensite, B = bainite BL = lower bainite, P = pearlite AR = residual austenite, F = ferrite, A = austenite, BU= upper bainite; M = martensite
(a) typical of FCC crystal	(b) coherent interface	(c) its a defective stacking of planes	(d) can be remove by annealing	
2. Grain refinement can be achieved using different methods, mark the clause which is NOT TRUE				
(a) both heat/cool rapidly	(b) GB pinning by precipitates	(c) heat slow and cool rapidly t	(d) recrystallize and cool fast	
3. Factors affecting formability- mark the clause which is NOT TRUE				
(a) high temperature	(b) strain rate	(c) small grain	(d) large grain	
4. What is the common factor to all strengthening methods in metals				
(a) presence of edge dislocations	(b) defects concentration	(c) dislocation density	(d) stress conditions	
5. The symbol name of Q-temperature in the figure				
(a) A1	(b) A3	(c) Acm	(d) A4 °C	
6. The domain name of R-region in the figure				
(a) F+A	(b) A+P	(c) Cementite+A	(d) Cementite+F	
7. A ₁ point decreases with increasing of: (mark the UNTRUE clause)				
(a) undercooling	(b) alloying	(c) cooling rate	(d) dislocation density	
8. Apply the lever rule for the austenite fraction to C ₀ alloy (figure)				
(a) ~5.8/7 %	(b) 5.8/6.2 %	(c) ~0.4/6.2 %	(d) ~0.4/7 %	
9. The symbol name of C-temperature in the TTT diagram (figure)				
(a) A2	(b) A3	(c) A1	(d) Acm	
10. The R-domain name in the TTT diagram (figure)				
(a) BL-TM+AR	(b) BL-TM	(c) Cementite+AR	(d) BL+AR	
11. How does Co alloying affect the TTT curves (figure)				
(a) Shift M ₉₀ down	(b) Shift M ₉₀ up	(c) Shift left	(d) Shift right	
12. To enhance wear resistance of a tempered AISI 1038 steel it must follow a process of:				
(a) flame hardening	(b) nitriding	(c) carburizing	(d) spheroidization	
13. Estimate the weight fraction of cooling products after transformation at point 2 as shown in the TTT diagram (see figure)				
a) 30%F+35%TM-BL+35%AR	b) ~50%F+ 40% BL-TM + 10%AR	c) 50%F+50 %TM-BL	d) 50%F + 50%AR	
14. Which process does result in a tensile residual stress state				
a) quenching in cold bath	b) cold rolling with small rolls	c) welding	d) induction surface hardening	
15. Which is the effective reducing agent of Iron ores in blast furnace for pig iron refinement				
(a) Carbon coke (b) Limestones (c) CO (d) CO ₂				
16. Which precipitation process is more effective for strengthening Aluminum alloys				
(a) natural aging (b) artifical aging (c) overaging (d) precipitation at GB				
17. Which of the following element in steels is not a deoxidizer element:				
(a) Si (b) Ni (c) Mn (d) Al				
18. Sensitization in austenitic stainless steels is caused by:				
(a) Impurities (b) GB precipitation of N (c) GB precipitation of Cr (d) Interstitial corrosion				

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 - (a) Impurities
 - (b) GB precipitation of N
 - (c) GB precipitation of Cr
 - (d) Interstitial corrosion

IMPORTANT TO AVOID CONFUSION IN OPTION CANCELLATIONS, FIRST **USE A PENCIL** OR ALTERNATIVELY, **MARK WITH A DOT** YOUR OPTION AND THEN **FILL THE FINAL OPTION LETTER** OF YOUR CHOICE WITH YOUR **PEN**. IN THE LIMIT A CROSS MARK (X) MEANS **OPTION CANCELLATION**!

1-d

dislocation can be removed by annealing not stacking fault lattice

2. Grain refinement can be achieved using different methods, mark the clause which is NOT TRUE (a) both heat/ cool rapidly (b) GB pinning by precipitates (c) heat slow and cool rapidly (d) recrystallize and cool fast

(c), heat slowly and cool rapidly.

Grain refinement is the process of reducing the size of the grains in a metal. This can be achieved by different methods, including:

- Heating the metal to a high temperature and then cooling it rapidly. This causes the grains to become smaller.
- Adding alloying elements to the metal. These alloying elements can act as grain refiners, which means that they can help to reduce the size of the grains.
- Pinning the grain boundaries. This can be done by adding precipitates to the metal. The precipitates will act as obstacles to the growth of the grains, which will help to reduce their size.

The option (c), heat slowly and cool rapidly is not a valid method of grain refinement. If the metal is heated slowly, the grains will have time to grow large. When the metal is then cooled rapidly, the grains will not have time to shrink significantly.

3. Factors affecting formability- mark the clause which is NOT TRUE : a) high temperature (b) strain rate (c) small grain (d) large grain

The answer is (d), large grain.

The formability of a metal is its ability to be shaped without breaking. The factors that affect the formability of a metal include:

- Temperature: The higher the temperature, the more ductile the metal will be. This means that it will be easier to deform without breaking.
- Strain rate: The strain rate is the rate at which the metal is deformed. The higher the strain rate, the less ductile the metal will be. This means that it will be more likely to break when it is deformed quickly.
- Grain size: The smaller the grain size, the more ductile the metal will be. This is because small grains have more grain boundaries, which act as obstacles to the movement of dislocations. Dislocations are defects in the crystal structure of a metal that can cause it to break.

So the correct answer is (d). Large grains are less ductile than small grains, so they will be more likely to break when they are deformed.

Here is a table that summarizes the effects of the different factors on the formability of a metal:

Factor	Effect
Temperature	Higher temperature → more ductile
Strain rate	Higher strain rate → less ductile
Grain size	Smaller grain size → more ductile

Factor	Effect on formability
Temperature	Increase in ductility, decrease in yield strength and strain hardening rate
Strain rate	Increase in strain hardening rate, decrease in ductility
Grain size	Increase in ductility, decrease in yield strength and strain hardening rate
Material properties	Yield strength, tensile strength, ductility, anisotropy
Process parameters	Die design, lubrication, blank holder force

Table 2

4. What is the common factor to all strengthening methods in metals: a) presence of edge dislocations (b)

defects concentration (C) dislocation density (D) stress condition

The answer is (C). Dislocation density.

All strengthening methods in metals work by increasing the dislocation density in the material. Dislocations are line defects in the crystal structure of a material. They can move through the crystal under an applied stress, and their movement causes the material to deform.

The higher the dislocation density, the more difficult it is for dislocations to move, which makes the material stronger.

The other options are not correct. Edge dislocations are a type of dislocation, but they are not the only type. Defects concentration is the presence of a high concentration of defects in a material, but it is not specific to strengthening methods. Stress condition is the state of stress in a material, but it is not a strengthening method.

Here are some of the strengthening methods that work by increasing dislocation density:

- Cold working: This involves plastically deforming the material at a low temperature. This increases the dislocation density by forcing dislocations to move through the crystal structure.
- Strain hardening: This is the process of increasing the strength of a material by plastically deforming it. This also increases the dislocation density.
- Solid solution strengthening: This involves adding impurities to the material. The impurities act as obstacles to dislocation movement, which increases the dislocation density.
- Precipitation hardening: This involves forming precipitates in the material. The precipitates act as obstacles to dislocation movement, which increases the dislocation density.

5-c Ac_m

6-c austenite + cementite (Fe₃C)

7-d

8-b

9-b the steel 1038 have 0.38% carbon "hypo-eutectoid" (A₃-A₁)

10-a

11-d

since we had a big amount of carbon from our alloy that 1.2% so the ttt diagraeme will shift to the right (The transformation start times for pearlite and bainite are delayed, meaning these transformations start at longer times and at lower temperatures during cooling.

The nose of the curve, which represents the fastest transformation time for pearlite, moves to longer times, which means the steel can be cooled more slowly before these transformations begin.

The overall effect of increasing the carbon content to 1.2% would be to make the steel more hardenable, allowing for a slower quench rate to achieve a given hardness and microstructure compared to a steel with lower carbon content like AISI 1038. This would generally be represented by a rightward shift of the TTT curve. This makes it easier to produce martensite and harder to produce pearlite and bainite during quenching.)

12-b (nitriding of austenitic stainless steel, expanded austenite is formed, which is responsible for the improved wear and corrosion resistance

Nitriding is commonly used with low carbon steels such as AISI 1038

Nitriding usually better than carburising (strengthening the surface) (surface heat treatment))

tempered steel --> nitriding

carburizing steel --> carburizing then quenching

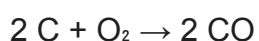
13-d

14-a

15-c

The primary and most effective reducing agent in blast furnaces for pig iron refinement is carbon monoxide (CO).

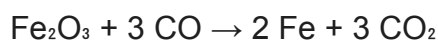
While coke (a) is indeed charged into the furnace and plays a crucial role, it primarily serves as a source of heat and carbon that react with oxygen to generate CO:



This exothermic reaction provides the high temperatures (around 2000°C) necessary for the reduction process.

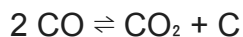
Here's a breakdown of the key players in a blast furnace:

- Coke: Supplies carbon and generates heat through combustion.
- Hot air blast: Increases reaction rates and creates turbulence for better material mixing.
- Flux (limestone): Removes impurities like silica and alumina from the molten iron, forming slag.
- Carbon monoxide (CO): The main reducing agent, reacting with iron oxide in the ore to yield metallic iron and carbon dioxide:



Reasons why CO is the effective reducing agent:

- Reactivity: At the high temperatures within the furnace, CO is more reactive than solid carbon for reducing iron oxide.
- Availability: The Boudouard equilibrium ensures adequate CO production within the furnace:



- Efficiency: CO reduction results in less CO₂ generation compared to direct use of carbon, reducing carbon emissions.

While other options mentioned have roles:

- Limestone (b): Acts as a flux, not a reducing agent.
- Carbon (c): Contributes to CO generation but is less reactive than CO itself.
- Carbon dioxide (d): Does not function as a reducing agent; instead, it's a product of the reduction process.

In conclusion, remember that carbon monoxide (CO) is the champion of reduction in blast furnaces, playing a vital role in transforming iron ore into pig iron.

16-d

Process	Temperature	Rate	Precipitate size	Strength
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Natural aging	Room temperature	Slow	Large and few	Low
Artificial aging	Elevated temperature	Controlled	Smaller and more uniform	Moderate
Overaging	Elevated temperature	Fast	Large and fewer	Low
Precipitation at GB	Elevated temperature	Controlled	Fine and dense	High

17-b

Here's why:

- Silicon (Si), Manganese (Mn), and Aluminum (Al) are all effective deoxidizers in steels. They react with dissolved oxygen in the molten steel to form stable oxides that float to the surface and are removed as slag. This process reduces the oxygen content in the steel, improving its properties like ductility, weldability, and resistance to corrosion.
- Nickel (Ni), on the other hand, primarily functions as an alloying element in steels. While it has an affinity for oxygen, its interaction with oxygen is not strong enough to effectively remove it from the molten steel.

18-d

Sensitization of Austenitic Stainless Steels

In austenitic stainless steels, sensitization refers to a phenomenon where specific heat treatments cause the depletion of chromium (Cr) near grain boundaries (GBs), making the steel susceptible to certain types of corrosion, particularly intergranular corrosion (IGC) and intergranular stress corrosion cracking (IGSCC)

Interstitial corrosion is a type of localised corrosion that occurs **at the grain boundaries** of metals and alloys. It is caused by the **penetration of corrosive agents** into the **interstices** (gaps) between the grains of the metal.

These gaps can be created by impurities, dislocations, or other defects in the crystal structure of the metal.

WRITTEN EXAM OF TMM

1. In the F-r Condon-Morse function the Young modulus is its:

(a) slope	(b) MIN	(c) MAX	(d) inflection point
-----------	---------	---------	----------------------
2. Which is the easy-to-slip crystal structure among the following?

(a) BCC	(b) FCC	(c) BCT	(d) HCP
---------	---------	---------	---------
3. Fe atom diffuses faster in BCC than in FCC crystal because:

(a) larger n. of interstitial sites	(b) smaller size of interstitials sites	(c) only at high temperature	(d) less defects
-------------------------------------	---	------------------------------	------------------
4. Select the reaction which DOES NOT occur in the blast furnace:

(a) $C + O_2 \rightarrow CO_2$	(b) $CO_2 + C \rightarrow 2CO$	(c) $CaCO_3 \rightarrow CaO + CO_2$	(d) none
--------------------------------	--------------------------------	-------------------------------------	----------
5. Which is the high temperature invariant reaction in C>2% steels

(a) eutectic	(b) eutectoid	(c) peritectic	(d) peritectoid
--------------	---------------	----------------	-----------------
6. In the C₁-alloy (figure) 100%M can be formed after equilibrium cooling to:

(a) A ₁ then quenching	(b) A ₃ then quenching	(c) Acm then quenching	(d) none
-----------------------------------	-----------------------------------	------------------------	----------
7. A₁ point (see figure) decreases its value with the alloying of:

(a) Si	(b) Mo	(c) Ni	(d) Ti
--------	--------	--------	--------
8. Estimate the cementite proportion in P at Co and To (see figure)

(a) 0.4-6.6/7%	(b) 0.4/7%	(c) 0.5%	(d) 0.5-6.6/7%
----------------	------------	----------	----------------
9. Give the symbol name for C temperature and R region in figure

(a) A ₁ /M+AR	(b) A ₃ /M+AR	(c) A ₃ /M	(d) Acm/M
--------------------------	--------------------------	-----------------------	-----------
10. What is the %C in AISI 4140 steel

(a) 4%	(b) 0.4%	(c) 0.41%	(d) 40%
--------	----------	-----------	---------
11. How does growth rate change with increasing temperature in figure

(a) decreases	(b) increases	(c) unchanged	(d) uncertain
---------------	---------------	---------------	---------------
12. What is the cause for the bainitic bay in the TTT diagram (figure)

(a) presence of Ni	(b) presence of large amount of C	(c) presence of Ni, Cr, Mo	(d) presence of Cr, Mo
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13. Given the TTT diagram (in figure), estimate the fraction of cooling products at point 2

(a) 75%BU + 25% AR	(b) 75% BU + 45% TM-BL + 5%AR	(c) 75%BU + 12.5% M+12.5% AR	(d) 75% BL +25% AR
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14. It is FALSE that TMCP structural steels exhibit:

(a) high Y _s	(b) high UTS	(c) high toughness	(d) low resilience
-------------------------	--------------	--------------------	--------------------
15. It is FALSE that Extensive solid solution can be achieved with:

(a) very low activation energy	(b) interstitials	(c) FCC structure	(d) large difference in Electronegativity
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16. Corrosion occurs preferentially at the cathode because:

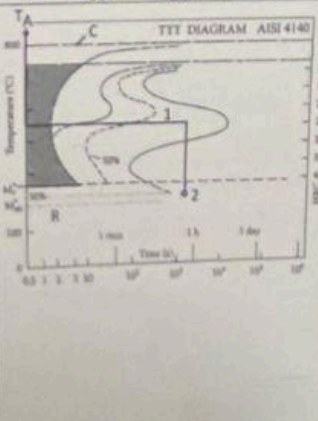
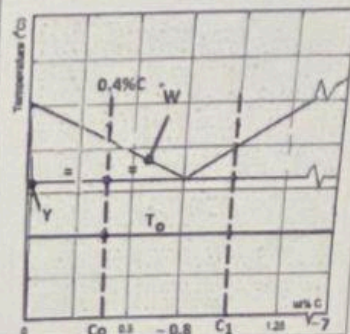
(a) e ⁻ accumulate here	(b) e ⁻ depart from here	(c) none	(d) H gas develops
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17. An AISI 1010 contains considerable amount of

(a) C	(b) Ni	(c) Cr	(d) none
-------	--------	--------	----------
18. PREN number for SS must be

(a) >15 for Austenitic	(b) the lower the better	(c) not applied to Austenitic	(d) <15 for Austenitic
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Legend of Symbols⁽⁶⁾:

TM = tempered martensite, B = bainite
 BL = lower bainite, P = pearlite
 AR = residual austenite, F = ferrite,
 A = austenite, BU= upper bainite;
 M = martensite, ppt = precipitates
 SS = stainless steels, Y_s = yield strength
 Q = activation energy for diffusion
 + division operator
 / separator



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 ALTERNATIVELY, MARK WITH A DOT YOUR OPTION AND THEN FILL THE FINAL OPTION LETTER OF
 YOUR CHOICE WITH YOUR PEN. IN THE LIMIT A CROSS-MARKED OPTION IS CONSIDERED INVALID.

1-a Franck–Condon principle energy diagram is energy / length

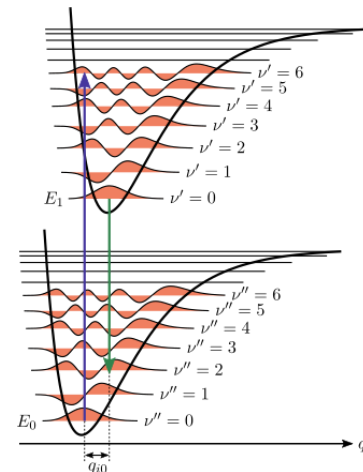
Young's modulus E is mathematically expressed as:

$E = \text{stress} / \text{strain}$

where:

- Stress is the force applied per unit area (F/A)
- Strain is the proportional change in length.

which makes the Young modulus the slope for this curve (function)



2-b(fcc)(high number of atoms → high ability to slip)

3-a

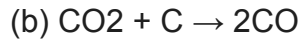
Explanation:

- (a) Larger number of interstitial sites: This is the correct answer. Fe atoms can diffuse through a crystal lattice by hopping from one interstitial site to another. In a bcc lattice, there are more interstitial sites per unit volume than in an fcc lattice. This means that there are more opportunities for Fe atoms to hop, and so diffusion is faster.
- (b) Smaller size of interstitial sites: This is incorrect. The size of the interstitial sites does not affect the diffusion rate as much as the number of interstitial sites. The bcc and fcc lattices have the same type of interstitial sites (tetrahedral and octahedral). The difference in diffusion rate is primarily due to the higher density of interstitial sites in the bcc lattice.
- (c) Only at high temperature: This is also incorrect. Diffusion can occur at any temperature, but it is faster at higher temperatures. The higher temperature increases the average kinetic energy of the atoms, which makes it easier for them to overcome the energy barrier required to hop between interstitial sites.
- (d) Less defects: This is incorrect. Defects can actually increase the diffusion rate by providing additional pathways for Fe atoms to move. Defects such as vacancies and self-interstitials provide more space for Fe atoms to move through, and they can also act as nucleation sites for new interstitial sites.

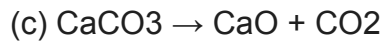
4-d

(a) $C + O_2 \rightarrow CO_2$

- Justification: This reaction is the combustion of carbon with oxygen, which is an essential step in the blast furnace process. It is used to provide the heat and energy necessary to reduce iron oxide to iron.



- Justification: This reaction is called the Boudouard reaction, which is a reversible reaction that occurs in the blast furnace. It produces carbon monoxide (CO), which is a valuable intermediate in the blast furnace process. The Boudouard reaction helps to control the amount of CO in the blast furnace, which is important for efficient iron production



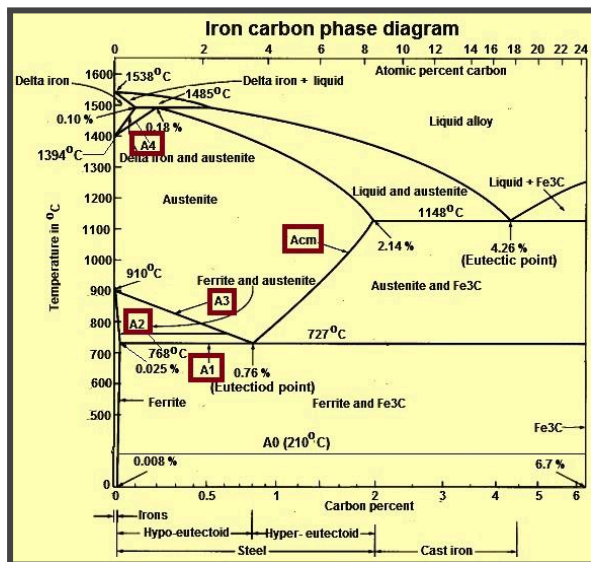
- Justification: This reaction is called the decomposition of calcium carbonate, which is an essential step in the blast furnace process. It produces calcium oxide (CaO), which is used as a flux to remove impurities from the molten iron. The decomposition of calcium carbonate provides the necessary carbon dioxide (CO₂) for the Boudouard reaction.

5-a

6-c(because when it headed above ACM (phase diagram) the microstructure of the alloy will be 100% austenite and we know that the austenite is transform to martensite so after quenching (heated and then rapidly cooled) it will form martensite after maintained for enough time

7-c

8-b



9-b

10-b

1. AISI: This stands for the American Iron and Steel Institute, which is the organization that developed the standard for steel grades in the United States.
2. 4140: This is the grade number of the steel. The grade number is a four-digit number that indicates the carbon content, alloying elements, and other properties of the steel. In the case of AISI 4140, the carbon content is 0.40%, and the alloying elements are manganese, chromium, nickel, and molybdenum.

11-b

In a Time-Temperature-Transformation (TTT) diagram, the growth rate of microstructure transformations typically increases with increasing temperature. This is because diffusion, the driving force for these transformations, becomes more rapid at higher temperatures. As a result, the transformation front can move more quickly through the material. However, there may be a minimum transformation temperature below which the transformation rate is very slow. This minimum temperature is often associated with a metastable phase or microstructure. The TTT diagram for a particular alloy will show the critical temperatures for different microstructure transformations at different temperatures. By understanding these critical temperatures, it is possible to control the microstructure of the alloy by controlling the cooling or heating rate.

12-d STEEL AISI 4140 contains Cr , Mo

13-a

14-d

15-d

16-a

Here is a summary of how corrosion occurs in metals in electron perspective:

- Oxidation reaction: Metal loses electrons to electrolyte.
- Electrons travel through external circuit to cathode.
- Reduction reaction: Oxygen is reduced to hydroxide ions by gaining electrons.
- Accumulation of electrons at cathode attracts metal ions from anode.
- Metal ions react with hydroxide ions to form metal hydroxide (corrosion product).

17-c

18-a

the PREN number for stainless steel (SS) must be greater than 15 for it to be considered austenitic. This is because austenite is the stable phase of stainless steel

at temperatures above 1,394°F (750°C). Stainless steel with a PREN number greater than 15 will retain its austenitic phase even at room temperature, which gives it its characteristic corrosion resistance.

WRITTEN EXAM OF TMM

- Which is the fastest diffusion process at low temperature
- (a) bulk diffusion (b) surface diffusion (c) Grain boundary diffusion (d) interstitial diffusion
2. Which is the final function of VOD in the post treatment of steel
- (a) decarburize (b) deoxidize (c) carburizing (d) degassing
3. By which factor is grain boundary energy influenced?
- (a) polymorphism (b) crystal anisotropy (c) misorientation angle (d) crystal structure
4. What does Electronegativity determine in metals
- (a) ionic bondic (b) nature of atomic bonds (c) electron motion (d) lattice defects interaction
5. How much do you expect in the proportion of austenite at T_{amb} when 5%Ti is alloyed to Co-alloy (see figure)
- (a) nearly 100% (b) very small (c) unrelated (d) a new phase grow up
6. Indicate the phase composition (Q/W, R) of microstructure (figure)
- (a) pearlite/ α , γ (b) α /cem, γ (c) γ /cem, α (d) α / γ , cem
7. Detect the %C in austenite in the Co-alloy at T_o (see figure)
- (a) $< 0.022\%C$ (b) 0.77% (c) 0.2% (d) 0.53%
8. Estimate the pearlite proportion of the Co-alloy at T_o (see figure)
- (a) 0.2/0.77% (b) ~ 0.2 / 0.5% (c) ~ 0.3 / 0.5% (d) 0.3/0.77%
9. How can temper embrittlement be prevented in steel processing
- (a) adding Mo (b) adding Cr (c) Adding Mn (d) Adding Si
10. Indicate the R-field name and the C temperature in Figure
- (a) metastable A/A_{cm} (b) metastable A/A_s (c) stable A/A_s (d) stable A/A_s
11. How does nucleation rate below B nose change with decreasing temperature in the TTT diagram (see figure)
- (a) increases (b) decreases (c) unaffected (d) undefined
12. For which steel, among AISI 1020, 4140, 4340, weldability is largest
- (a) AISI 1020 (b) AISI 4140 (c) AISI 4340 (d) none
13. Estimate the fraction of cooling products down to point 2
- (a) 50% BU +50% AR (b) 50% M +50% TM-BL (c) 50% M +50% AR (d) 50% TM-BL +50% AR
14. Which is the most suitable application for AISI 1095 steel?
- (a) ignition valve (b) crankshaft (c) punching tool (d) stamping panel
15. How does cold rolling enhance fatigue resistance of long rolled products
- (a) severe strain hardening (b) using small rolls (c) controlled rolling (d) on-
16. Which of the following factors DOES NOT influence galvanic corrosion
- (a) like metals in contact (b) surface area ratio of electrodes (c) pH (d) hum
17. Determine the %C and %Cr in the X40Cr14 steel
- (a) 4% C / 14% Cr (b) 0.4% C / 14% Cr (c) 40% C / 3.5% C (d) 40% C / 14% Cr
18. Which of the following metals IS NOT corrosion protective of metals
- (a) Ti (b) Zn (c) Mg (d) W

FAIL

IMPORTANT TO AVOID CONFUSION WITH POSSIBLE CANCELLATION
ALTERNATIVELY, MARK WITH A DOT YOUR OPTION AND THEN
YOUR CHOICE WITH YOUR PEN. IN THE LIMIT A CROSS-MARK

1-b

b) surface diffusion.

Surface diffusion is the fastest diffusion process at low temperatures because it does not require the movement of atoms through the bulk of the material. At low temperatures, there are very few vacancies available, so the process of vacancy diffusion is very slow. However, surface diffusion can still occur because atoms can move along the surface of the material without having to jump over vacancies.

2-d

3-3

Grain boundary energy is the energy required to create a new grain boundary in a material. It is influenced by several factors, including the misorientation angle between the grains. The misorientation angle is the angle between the lattice planes of two adjacent grains. Higher misorientation angles generally result in higher grain boundary energy. This is because higher misorientation angles mean that the atoms at the grain boundary are less well-matched, which requires more energy to bring them together.

4-c

Electronegativity in metals determines their tendency to lose electrons in chemical bonding. Metals generally have low electronegativity, indicating a willingness to donate electrons.

In metals, electronegativity has minimal impact on bonding. Metals predominantly form metallic bonds, where electrons move freely throughout the lattice.

Electronegativity, which is crucial in ionic or covalent bonds, is less relevant in metals. The key role of electronegativity in metals lies in the mobility of electrons within the structure, contributing to properties like conductivity

5-b

6-

7-d

8-b

9-a

10-b

11-b

12-a, lower carbon steel means more weldability and AISI 1020, is the lowest carbon content steel with 0,2 %

13-c

14

15-b

16-a

17-b

18-a

1-b

WRITTEN EXAM OF TMM

1. It is FALSE on the advantages of pure metals vs. alloys that are:				Legend of Symbols⁽¹⁾: TM = tempered martensite, B = bainite BL = lower bainite, P = pearlite AR = residual austenite, F = ferrite, A = austenite, BU = upper bainite; M = martensite SS = stainless steels T-line = temperature line + division operator / separator
(a) > ductile	(b) > corrosion resistant	(c) tougher	(d) stiffer	
2. It is FALSE in Humé-Rothery rules that all provide indications for compound formation depending on				
(a) Electronegativity difference	(b) atom radii difference = 25%	(c) valence	(d) crystal structure	
3. What is formed when electronegativities of atoms differ?				
(a) intermetallic compound	(b) liquid solution	(c) solid solution	(d) alloy steels	
4. Precipitation hardening does NOT involve the following step:				
(a) solutionizing	(b) quenching	(c) aging	(d) phase change	
5. Determine %C in Y and the symbol of T-line in W (see figure)				
(a) 0.022%/A _{cm}	(b) 0.0025%/A _{cm}	(c) 0.0025%/A ₃	(d) 0.022%/A ₃	
6. How does A ₁ change after alloying 5% Mn to initial Co-alloy				
(a) decreased	(b) increased	(c) unchanged	(d) disappear	
7. For C ₁ -alloy in figure, which temperature must be reached under equilibrium before final quenching to form 5% cementite and 95% M				
(a) down to T _{amb}	(b) down to A _{cm}	(c) down to A ₁	(d) down to A ₃	
8. Estimate the proportion of pearlite of Co-alloy at T ₀				
(a) ~10 %	(b) ~40 %	(c) ~0.2+7 %	(d) 50 %	
9. Indicate the R-field name in the TTT diagram (figure)				
(a) F+P	(b) F+cementite	(c) stable A	(d) BU	
10. Indicate the T-line in C-field name in the TTT diagram (figure)				
(a) A ₃	(b) A ₃	(c) A _{cm}	(d) A ₁	
11. How is microstructure toughness affected in the AISI 4140 steel with decreasing temperature in the TTT diagram (see figure)				
(a) increased	(b) decreased	(c) unaffected	(d) undefined-	
12. Which of the following alloys may contain some pro-eutectoid cementite upon slow cooling				
(a) AISI 1020	(b) AISI 4340	(c) AISI 1095	(d) AISI 304	
13. Estimate the proportion of cooling products on cooling down to point 2 (see figure)				
a) 50% M + 50 % TM-BL	b) 50% M + 45% TM-BL + 5%AR	c) 50% TM-BL + 50% AR	d) 90% TM-BL + 10%AR	
14. What is the more suited class of steels for induction hardening?				
a) austenitic SS	b) tempering steel	c) low carbon steel	d) None	
15. In which alloy strain hardening is most effective for highest strength?				
(a) Aluminum alloys	(b) Co-alloy	(c) Carbon steels	(d) Mg alloys	
16. How is the eutectic composed in the Al-Si alloy system?				
(a) Al + Si	(b) α (solid solution) + β (solid solution)	(c) Si+liquid	(d) α (solid solution) + Si	
17. Which are the precipitates formed since early aging in AlXXX alloys?				
(a) Al ₂ Cu GP zones	(b) Al ₂ Mn GP zones	(c) Al ₂ Mn ₃ GP zones	(d) Al ₂ Mg GP zones	
18. In which metal the rate of corrosion is highest?				
(a) Cu	(b) Fe	(c) Zn	(d) Ti	

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1- b

alloys are less ductile than metals because they are made of 2 or more types of atoms

alloys are tougher and stiffer than pure metals

alloys are corrosion resistant

2- The Hume-Rothery rules are a great way to predict if two elements will form a solid solution. These rules state that to be able to form a solid solution, the two elements must have similar atomic sizes, crystal structures, electronegativities, and valencies

The atomic radii of the solute and solvent atoms must differ by no more than 15%:

b

3-a

Intermetallic compounds:

- Occur when there is a significant difference in electronegativity between the atoms involved. This difference creates a strong ionic or covalent bond between the atoms, leading to a distinct crystalline structure unique to the compound.
- Examples of intermetallic compounds include NiTi (used in shape-memory alloys) and CuZn (brass).

5-a

6-a

A1 decrease with increasing of Mn or Ni

7-c

after calculation the % of cementite was 3%

$(1-0.8)/(7-0.8)$

8-d

$(0.4-0.022)/(0.8-0.022)=50$

9-c (stable austenite)

10- (d)a-1

11) b)

****Tempering Temperature:**

- Higher tempering temperatures lead to a microstructure with tempered martensite and retained austenite, enhancing toughness.

- Lower tempering temperatures may increase hardness but could potentially reduce toughness.

****Impact Transition Temperature:** Decreasing temperature raises the impact transition temperature, making AISI 4140 steel more prone to brittle fracture at lower temperatures and potentially reducing toughness.

****Martensite Influence:** Lower temperatures accentuate the brittleness of martensite, impacting the overall toughness of the steel.

12-a 1020 (0.2% carbon)

13-